

Information Retrieval #7 – Ranked Retrieval

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17.04.2026



Inhaltsverzeichnis/Agenda

1. Bonus Quiz Tricks
2. Lecture Recap
3. Exercise 6: Heap's and Zipf's Law
4. Kahoot

Exercise 5: Index Compression

Largest code points:

- Variable Length: Check packet size, remove first bit of every segment
- Fix-length: Size of maximum encoding
- Unary: size – 1
- Gamma: If n is the size, the largest number that fits in $\left\lceil \frac{n-1}{2} \right\rceil$

Exercise 5: Index Compression

Identifying Encoding Methods:

- Gamma Encoding: Always starts with string of 1s, middle bit is 0, always odd size
- Variable length encoding: Multiple of packet size, all packets except last should have 0 as leading bit
- Fixed-length encoding: Multiple of packet size, no deterministic leading
- Unary: String of 1s followed by a 0

Exercise 5: Index Compression

Ordering Encoding Methods:

- Gamma Encoding (n is number in decimal): $2 \cdot \lfloor \log_2 n \rfloor + 1$
 - Intuition:
 - $\lfloor \log_2 n \rfloor$ many 1s in unary
 - 1 bit for unary cutoff
 - $\lfloor \log_2 n \rfloor$ for actual binary representation (since we truncate the leading 1)
- Variable Length: If n = number in binary, p = packet size, then we have $\left\lceil \frac{|n|}{p-1} \right\rceil \cdot p$
- Unary: Value of number + 1
- Fixed-length encoding: Length of binary rounded up to packet size

Recap: Heap's and Zipf's Law

N: number of documents, M: number of terms, T: number of tokens

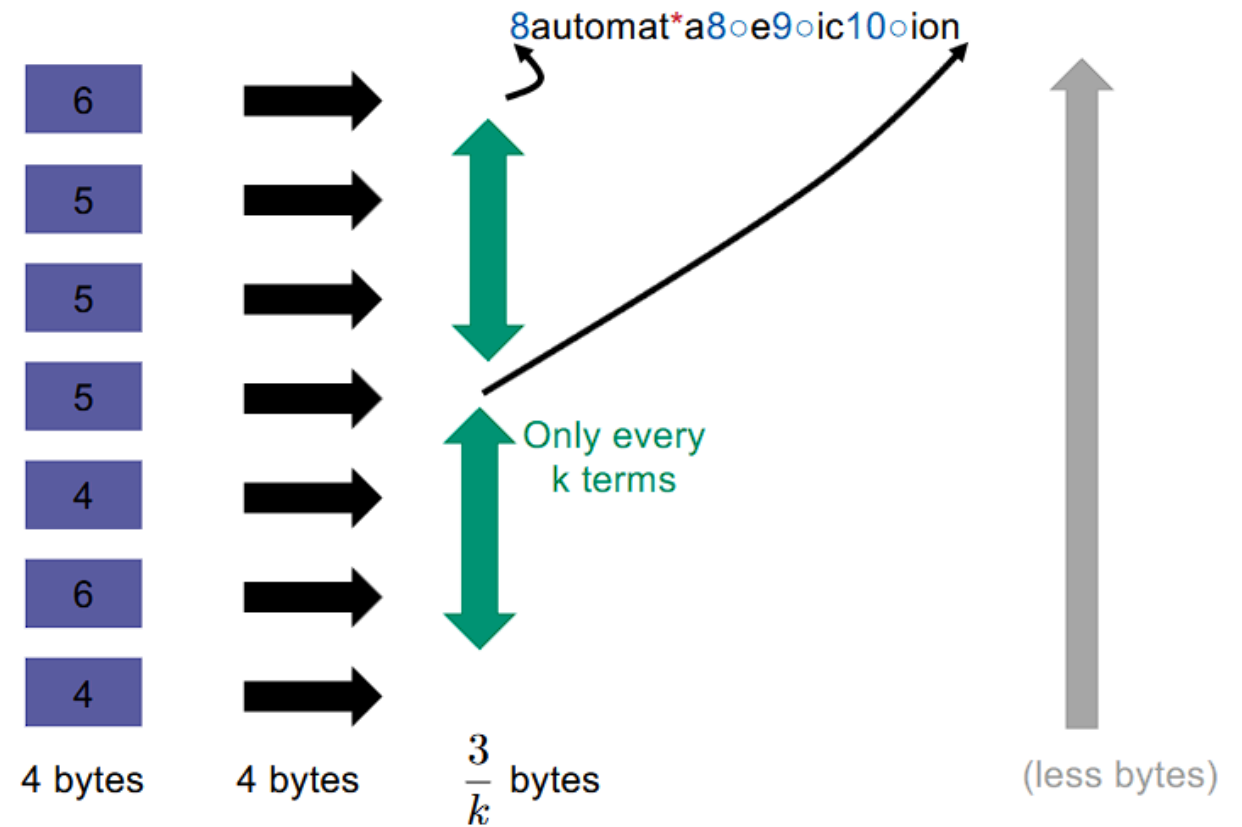
Heap's Law: $M = k\sqrt{T}$, empirically $30 \leq k \leq 100$.

Zipf's Law: Frequency = $\frac{k}{\text{Rank}}$

Recap: Compression Techniques

Compression of the dictionary:

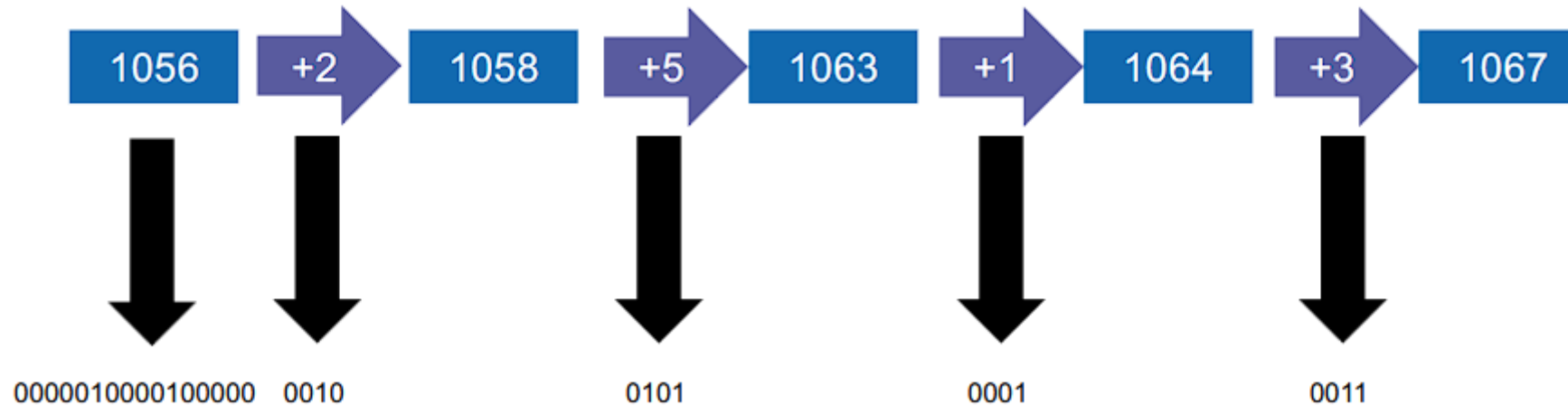
- Front Coding, Blocked Storage



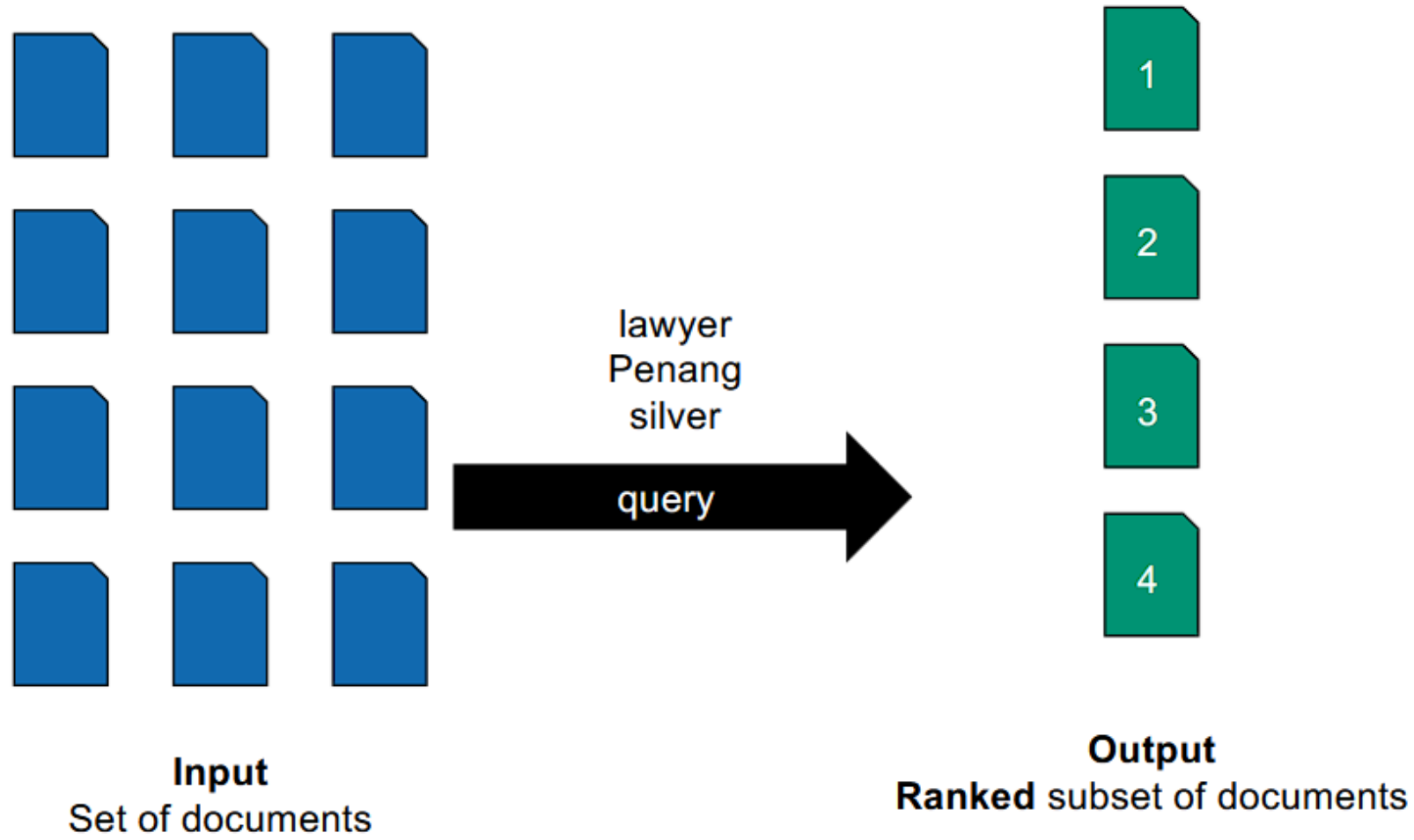
Recap: Compression Techniques

Compression of the postings lists:

- Encoding Gaps
 - Only works for frequent terms!

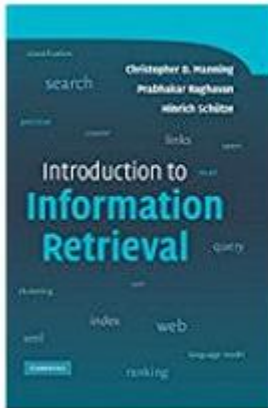


Ranked Retrieval



Parametric Search

- Split index up between metadata



Title: Introduction to Information Retrieval

Authors: Christopher D. Manning and Prabhakar Raghavan and Hinrich Schütze

Hardcover: 506 pages

Publisher: Cambridge University Press; 1 edition (July 7, 2008)

Language: English

ISBN-10: 0521865719

ISBN-13: 978-0521865715

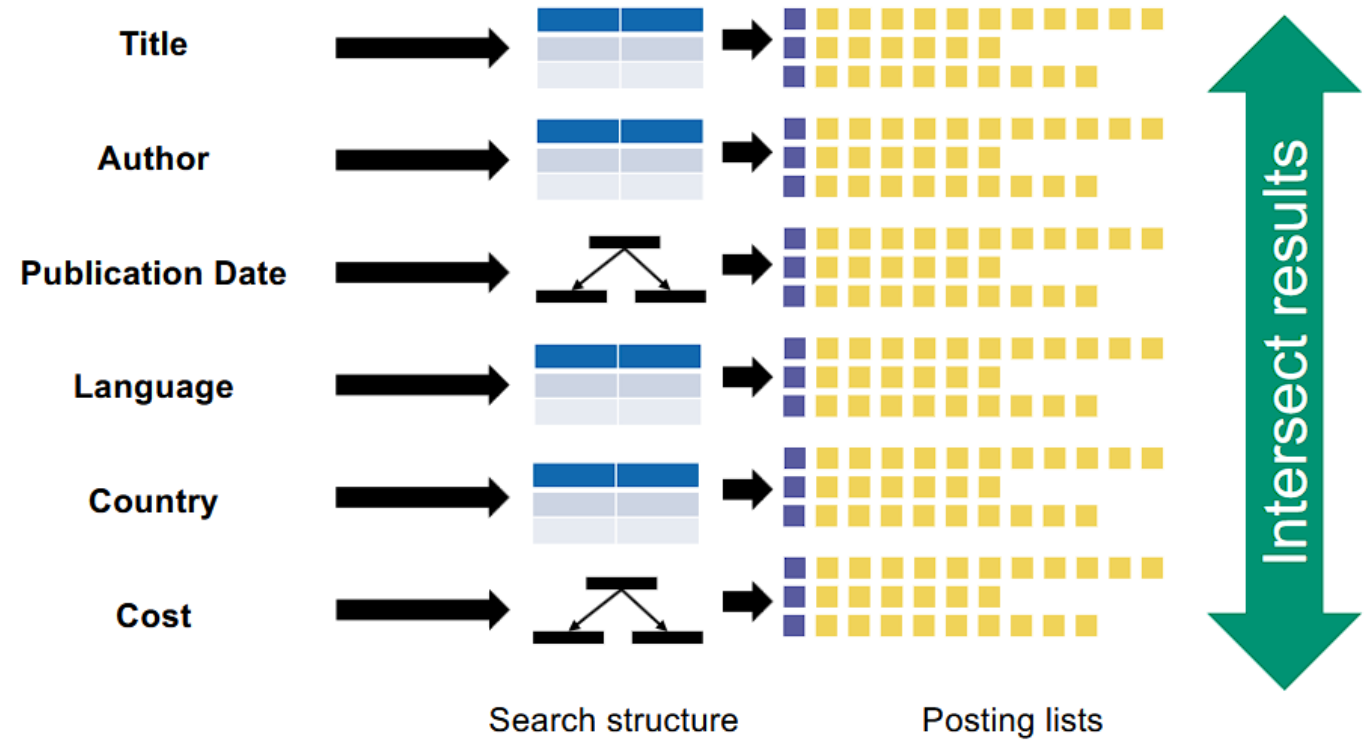
Product Dimensions: 7 x 1.2 x 10 inches

Shipping Weight: 2.2 pounds ([View shipping rates and policies](#))

Average Customer Review: ★★★★★ ⌵ [27 customer reviews](#)

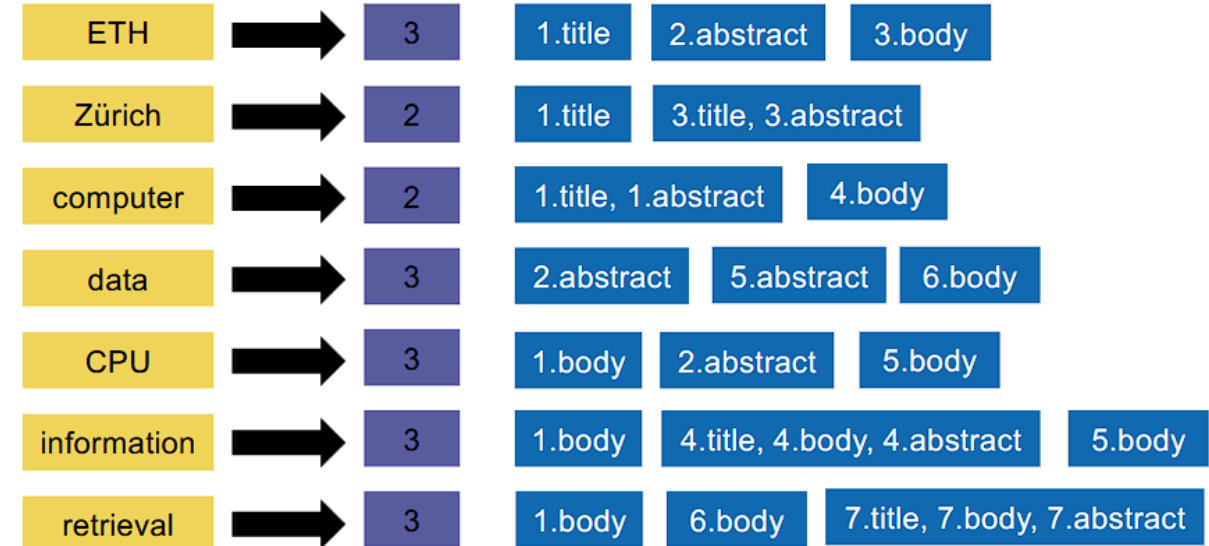
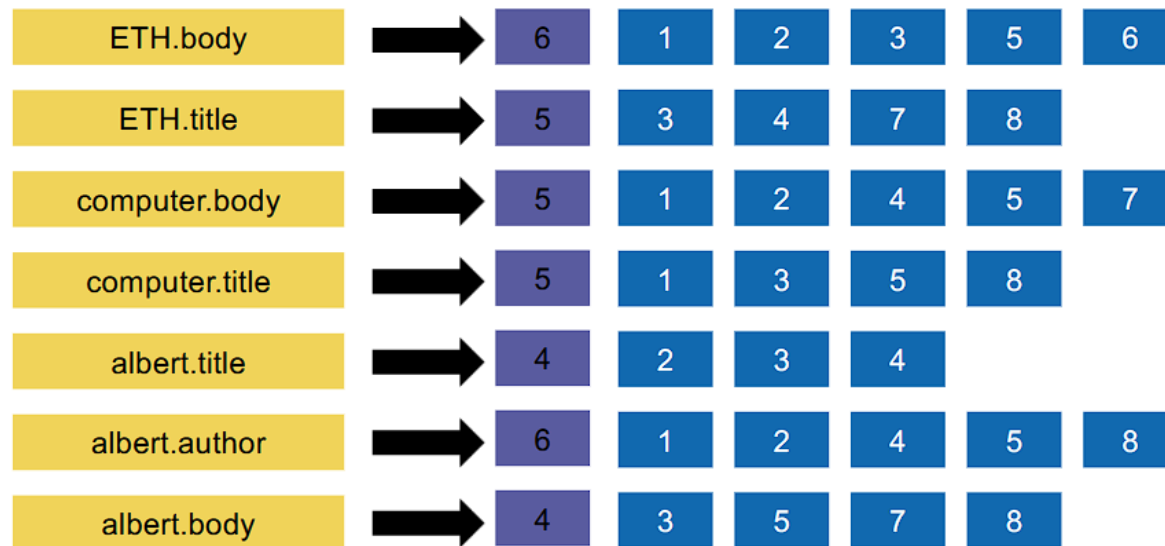
Parametric Search

- One index for each metadata entry
- Intersect results at the end
 - Zone search



Zone Search

- Zones in terms vs. Zones in postings



Scoring

- Single-term query
- Rank by total score

$$\sum_{i=1}^{i=l} g_i s_i$$

Boolean results
1 = zone i contains term
0 = zone i does not contain term

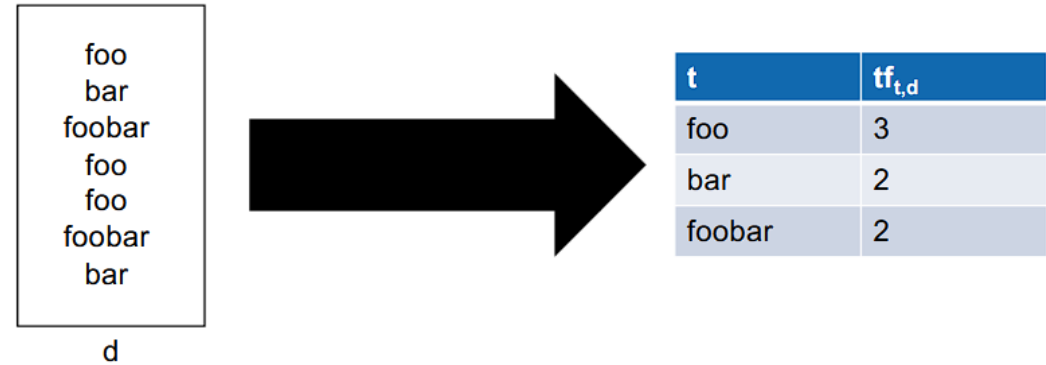


Zone	weight		weight
Title	0.3	→	0.3
Abstract	0.2		
Body	0.5	→	0.5

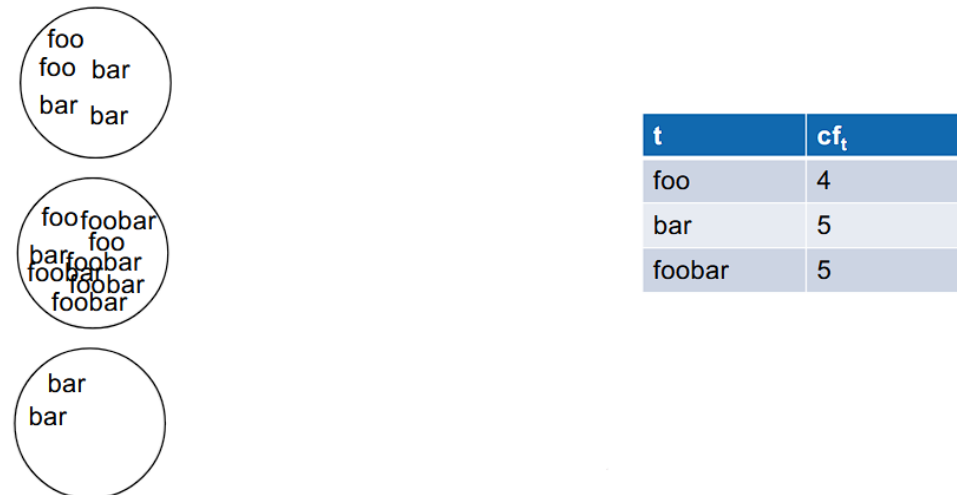
Total score	0.8
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Weights

- Term Frequency:

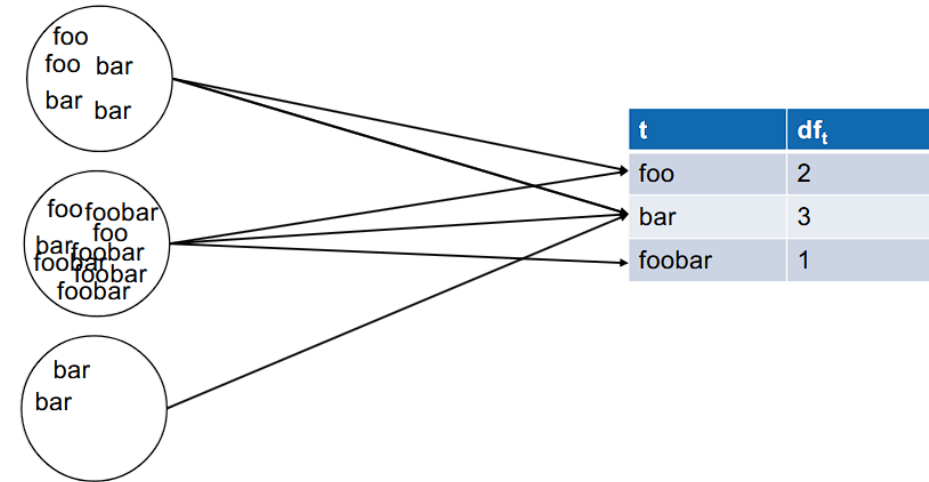


- Collection frequency:

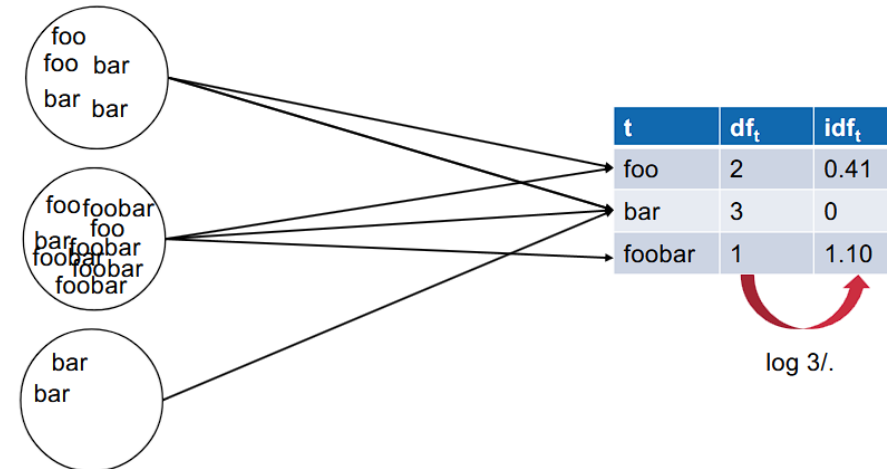


Weights

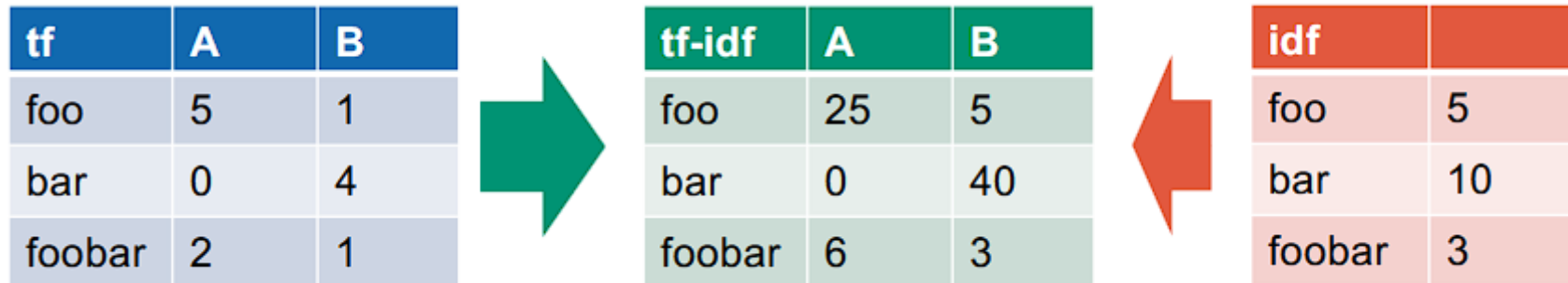
- Document Frequency:



- Inverse Document Frequency:
 - $idf_t = \log \frac{N}{df_t}$, where N: number of docs



tf-idf



Exercise 6: Heap's and Zipf's Law

- Moodle: Questions about lecture content
- Programming: Plot Heap's and Zipf's Law

Kahoot

<https://create.kahoot.it/share/information-retrieval-ranked-retrieval/4ce4ee62-9d8b-478c-b73f-8876e8ec77ef>